

Week 1: Requirement Analysis and Customer Database Module

1. Requirement Analysis

Understanding E-Commerce Business

An e-commerce system manages the end-to-end flow of buying, selling, and managing products online. Below is a structured breakdown of the business process, customer data requirements, and entity definitions for system design and database modeling.

The Process Flow:



- Browse / Search Product: The Customer Searches for the product they want to purchase.
- Add to Cart: After choosing the product needed, they will add it to cart.
- Order Placed: The customer chooses the address to deliver and place the order.
- Checkout/ Payment: After Placing the order, the system make them to move to payment section and then checkout process.
- Delivery: After payment, the product will be assigned to the seller for dispatch and the product will be delivered.
- Review & Support: After successful delivery, the customer may give review about the product. For some product, they might need support for installation or return or replace.

Entities:

Entity	Description	Key Attributes	Key Relationships
Customer	End-user who browses and purchases products.	cust_id (PK), first_nm, last_nm, email_address, phone_no, registration_date	One-to-Many with Order, Review
Seller	Merchant or vendor listing products on the platform.	seller_code (PK), company_title, contact_email, phone_number, vendor_rating, banking_details	One-to-Many with Product
Product	Item listed for sale on the platform.	prod_id (PK), seller_code (FK), prod_title, description_text, unit_price, available_qty, sku_code, category_id	Many-to-One with Seller, One-to-Many with Review
Order	Transaction instance representing a customer purchase.	order_id (PK), cust_id (FK), booking_date, order_status (pending, shipped, delivered, cancelled), gross_amount, delivery_address	Many-to-One with Customer, One-to-One with Payment, Many-to-Many with Product (via OrderItem)
Payment	Monetary transaction tied to an order.	payment_code (PK), order_id (FK), payment_channel (credit card, UPI, PayPal), txn_status (success, failed, refunded), reference_txn_id, total_paid, payment_timestamp	One-to-One with Order
Review	Feedback and rating submitted by a customer for a product.	feedback_id (PK), cust_id (FK), prod_id (FK), star_score (1–5), user_feedback, posted_date	Many-to-One with Customer, Many-to-One with Product

Customer Table Creation:

```
SQL> create table customer_info(cust_id varchar2(20) Primary key, first_nm varchar2(20),last_nm varchar2(20),email_ad
dress varchar2(30),phone_no number(10),registration_date date);

Table created.

SQL> desc customer_info
Name                               Null?    Type
-----
CUST_ID                           NOT NULL VARCHAR2(20)
FIRST_NM                           VARCHAR2(20)
LAST_NM                           VARCHAR2(20)
EMAIL_ADDRESS                      VARCHAR2(30)
PHONE_NO                           NUMBER(10)
REGISTRATION_DATE                  DATE
```

Insert Values:

```
SQL> INSERT INTO customer_info
2 VALUES ('CUST-501', 'Marcus', 'Vance', 'm.vance@apexmail.net', 9123456780, TO_DATE('2023-01-18', 'YYYY-MM-DD'));

1 row created.
```

```

SQL> INSERT INTO customer_info (cust_id, first_nm, last_nm, email_address, phone_no, registration_date)
2 VALUES (
3   '&cust_id',
4   '&first_nm',
5   '&last_nm',
6   '&email_address',
7   &phone_no,
8   DATE '&registration_date'
9 );
Enter value for cust_id: CUST-502
old 3:  '&cust_id',
new 3:  'CUST-502',
Enter value for first_nm: Elena
old 4:  '&first_nm',
new 4:  'Elena',
Enter value for last_nm: Rostova
old 5:  '&last_nm',
new 5:  'Rostova',
Enter value for email_address: elena.r@novanet.org
old 6:  '&email_address',
new 6:  'elena.r@novanet.org',
Enter value for phone_no: 9234567891
old 7:  &phone_no,
new 7:  9234567891,
Enter value for registration_date: 2024-08-14
old 8:  DATE '&registration_date'
new 8:  DATE '2024-08-14'

1 row created.

```

Like this we can create as many rows we want.

```

SQL> select * from customer_info
2 ;

```

CUST_ID	FIRST_NM	LAST_NM	EMAIL_ADDRESS	PHONE_NO	REGISTRAT
CUST-501	Marcus	Vance	m.vance@apexmail.net	9123456780	18-JAN-23
CUST-502	Elena	Rostova	elena.r@novanet.org	9234567891	14-AUG-24
CUST-503	Kavitha	Sundaram	kavitha.s@zenithtech.in	9345678902	05-OCT-23
CUST-504	David	Kowalski	dkowalski@bluestar.com	9456789013	12-MAR-25

Update:

```
SQL> update customer_info set phone_no = 9811223344 where cust_id = 'CUST-504';  
  
1 row updated.
```

3. Conclusion

In this laboratory session, the foundational database requirements for an e-commerce platform were established. The operational workflows were systematically mapped into corresponding data entities with well-defined relational structures. A dedicated customer profile table was implemented in Oracle SQL*Plus with appropriate data types and constraints. Both static and parameterized data entry methodologies were evaluated, followed by exhaustive validation through SELECT queries and targeted UPDATE operations to ensure data integrity.